



SVP-25-011

10 CFR 50.73

March 7, 2025

U.S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, D.C. 20555

Quad Cities Nuclear Power Station, Unit 1 and 2
Renewed Facility Operating License No. DPR-29 and DPR-30
NRC Docket No. 50-254 and 50-265

Subject: Licensee Event Report 254/2025-001-00 "Technical Specification 3.6.4.2 Action Not Performed Due to Tagout Boundary Placement Error"

Enclosed is Licensee Event Report 254/2025-001-00 "Technical Specification 3.6.4.2 Action Not Performed Due to Tagout Boundary Placement Error," for Quad Cities Nuclear Power Station, Unit 1.

This report is being submitted in accordance with 10 CFR 50.73(a)(2)(i)(B) for an operation or condition which was prohibited by the plant's Technical Specifications.

There are no regulatory commitments contained in this letter.

Should you have any questions concerning this report, please contact Conner Bealer at 779-231-6207.

Respectfully,

A handwritten signature in black ink that reads "Doug Hild".

Doug Hild
Site Vice President
Quad Cities Nuclear Power Station

cc: Regional Administrator – NRC Region III
NRC Senior Resident Inspector – Quad Cities Nuclear Power Station



LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)
(See NUREG-1022, R.3 for instruction and guidance for completing this form
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Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by email to Infocollections.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. Facility Name

Quad Cities Nuclear Power Station, Unit 1

☒ 050
☐ 052

2. Docket Number

00254

3. Page

1 OF 4

4. Title

Technical Specification 3.6.4.2 Action Not Performed Due to Tagout Boundary Placement Error

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved		
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name		Docket Number
01	16	2025	2025	- 001 -	00	03	07	2025	Quad Cities Nuclear Power Station Unit 2	☒ 050	00265
									Facility Name	☐ 052	Docket Number

9. Operating Mode

1 – Power Operations

10. Power Level

093

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)

☒ 10 CFR Part 20	☐ 20.2203(a)(2)(vi)	☒ 10 CFR Part 50	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.1200(a)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	☐ 73.1200(b)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	☐ 73.1200(c)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)	☐ 73.1200(d)
☐ 20.2203(a)(2)(i)	☒ 10 CFR Part 21	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	☒ 10 CFR Part 73	☐ 73.1200(e)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.77(a)(1)	☐ 73.1200(f)
☐ 20.2203(a)(2)(iii)		☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(2)(i)	☐ 73.1200(g)
☐ 20.2203(a)(2)(iv)		☒ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(ii)	☐ 73.1200(h)
☐ 20.2203(a)(2)(v)		☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)		

☐ OTHER (Specify here, in abstract, or NRC 366A).

12. Licensee Contact for this LER

Licensee Contact

Richard Swart, Principal Regulatory Engineer

Phone Number (Include area code)

309-227-2810

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS

14. Supplemental Report Expected

☒ No ☐ Yes (If yes, complete 15. Expected Submission Date)

15. Expected Submission Date

Month	Day	Year
N/A		

16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)

On January 16, 2025, at 1657 CDT, it was discovered that manual Secondary Containment Isolation Valve (SCIV) 2-1279-21A, 2A Reactor Water Clean Up Filter Demineralizer Rad Waste Drain Valve, was in an Open position. This resulted in discovery of a condition which was prohibited by Technical Specifications (TS). The valve should have been in a Closed position or should have had appropriate administrative controls applied when left in an Open position under the applicable tagout boundary that had been established. The valve was closed on January 16, 2025, at 1801 CDT in accordance with TS 3.6.4.2 Condition A. However, a past functionality review indicated the valve had been open for approximately 86 hours total, which exceeded TS 3.6.4.2 Condition A.1 to isolate the line in 8 hours.

The cause of the event was a failure to implement administrative controls for the SCIV left in an open position during tagout boundary dispatch and placement.

Immediate action was to close the SCIV. Follow up actions will be to improve SCIV tagout boundaries and valve labels.

This report is being submitted under 10 CFR 50.73(a)(2)(i)(B) for an operation or condition which was prohibited by the plant's Technical Specifications.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
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1. FACILITY NAME Quad Cities Nuclear Power Station Unit 1	☒ 050	2. DOCKET NUMBER 00254	3. LER NUMBER		
	☐ 052		YEAR 2025	SEQUENTIAL NUMBER - 001	REV NO. - 00

NARRATIVE**PLANT AND SYSTEM IDENTIFICATION**

General Electric - Boiling Water Reactor, 2957 Megawatts Thermal Rated Core Power

Energy Industry Identification System (EIIIS) codes are identified in the text as [XX].

EVENT IDENTIFICATION

Technical Specification 3.6.4.2 Action Not Performed Due to Tagout Boundary Placement Error

A. CONDITION PRIOR TO EVENT

Unit: 1 Event Date: January 16, 2025 Event Time: 1657 CDT
Reactor Mode: 1 Mode Name: Power Operation Power Level: 93%

There were no other structures, systems or components (SSC) inoperable during this event time period that could have contributed to this event.

B. DESCRIPTION OF EVENT

On January 16, 2025, at 1657 CDT, it was discovered that manual secondary containment isolation valve (SCIV)[ISV] 2-1279-21A, 2A Reactor Water Clean Up [CE] Filter Demineralizer Rad Waste Drain Valve, was in an Open position. Operations discovered this during maintenance activities within the tagout boundary that was to be provided by this valve. The valve was tagged in an Open position using a Caution/Information card to ensure the piping remained drained during the work on an adjacent valve. The 2-1279-21A could have been used as a drain for the volume, under applicable procedural and administrative controls, and then later Closed, although there were no specific instructions in the tagout to close the valve after draining. The tagout did include cautions and procedure references for the 2-1279-21A related to it being a SCIV.

When the Open position of the valve was discovered, Technical Specification (TS) 3.6.4.2 Condition A was entered. The 2-1279-21A valve was closed at 1801 CDT, and TS 3.6.4.2 Condition A was exited. A past functionality review indicated that the valve had been left in an Open position from when the tagout was validated at 0409 on January 13, 2025, until it was closed at 1801 CDT on January 16, 2025, for a total of 85 hours and 52 minutes. This exceeds the 8 hour completion time in TS 3.6.4.2 Condition A Action A.1 to isolate the secondary containment flow path. Issue Report 04830292 was generated.

A tagout package includes multiple sections, including placement instructions at the front of the package, and card notes at the end, in addition to the actual physical tags, called EPN (Equipment Part Numbers) Tags, that are hung on the equipment. Specific to this tagout, the EPN Tag had a note referring to a governing procedure for SCIV openings, and a note that secondary containment could be violated. These secondary containment impacts were not fully recognized by the operations team hanging the tagout, and the necessary administrative controls were not discussed in detail in a pre-job brief.

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NARRATIVE**C. CAUSE OF EVENT**

The cause of the self-identified event was a failure to implement administrative controls for the SCIV left in an open position as part of the tagout.

The Senior Reactor Operator responsible for dispatching the tagout was overly confident in their knowledge of the specific system and previous tagouts. They were preoccupied with a different tag that had expected alarms, which led to an incomplete review of all tagout documents. The Nuclear Station Operator responsible for dispatching the tagout reviewed all documents, but their lack of attention to detail led to a failure to recognize the potential TS implications. They also assumed that the valve would ultimately be left in a closed position. The Equipment Operators hanging the tagout did not question an abnormal note on the EPN tag for a valve they were leaving in the open position. Collectively, the individuals did not recognize the need to establish appropriate administrative controls from a procedure referenced in the tagout for valve 2-1279-21A being left in an open position.

Contributing were enhancement opportunities in the tagout. The secondary containment impacts could have been located in the front section of the tagout and could have been color coded for improved recognition. The reference to a procedure number for SCIVs could have been enhanced with a statement that the valve should not be left in an open position without an operator in attendance.

D. SAFETY ANALYSIS**System Design**

The function of the SCIVs, in combination with other accident mitigation systems, is to limit fission product release during and following postulated Design Basis Accidents (DBAs). Secondary containment isolation within the time limits specified for those isolation valves designed to close automatically ensures that fission products that leak from primary containment following a DBA, or that are released during certain operations when primary containment is not required to be operable or take place outside primary containment, are maintained within the secondary containment boundary.

The normally closed manual SCIVs are considered operable when the valves are closed and blind flanges are in place, or open under administrative controls.

Updated Final Safety Analysis Report (UFSAR) Section 6.2.3.1 provides that the safety objective of the secondary containment system, in conjunction with other engineered safeguards and nuclear safety systems, is to limit the release of radioactive materials so that offsite doses resulting from a postulated DBA will remain below 10 CFR 100 guideline values.

Both Units 1 and 2 share a common Reactor Building (secondary containment). However, there is a piping difference between the two QCNPS units. Specifically, Unit 2 valve 2-1279-21A is an SCIV; whereas, the equivalent valve on Unit 1 does not serve an SCIV function.

Safety Impact

There were no safety consequences as a result of this event. Although maintenance activities did result in an open flow path through this SCIV related pipe, the breach size was determined to be 7.1 square inches. Existing Engineering Evaluation EC 347977 concludes that any hole found in Secondary Containment which is less than 12.5 sq in would not

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	☐ 052		YEAR 2025	SEQUENTIAL NUMBER - 001	REV NO. - 00

NARRATIVE

render the secondary containment inoperable. Biennial secondary containment capability tests demonstrate proper differential pressures being maintained with a 4 inch diameter opening, supporting the EC Evaluation. At all times during this event, Standby Gas Treatment remained operable, and normal Reactor Building Ventilation maintained proper negative pressures on secondary containment. Thus, there was no release of radioactivity from secondary containment as a result of this event. This event is not considered a safety system functional failure.

E. CORRECTIVE ACTIONS

Immediate:

1. Closed SCIV 2-1279-21A.

Follow-up:

1. Improve the tagout for SCIVs to more prominently highlight the requirements for administrative controls any time the valve is left in an Open position.
2. Improve labeling of SCIVs to indicate that administrative controls are required when left open.
3. Removed qualifications of workers involved and performed remediations.
4. Performance Management has been applied to workers involved.

F. PREVIOUS OCCURRENCES

The station events database, LERs and IRIS were reviewed for similar events at Quad Cities Nuclear Power Station.

LER 254/2024-001-00 (01/25/2025), Technical Specification 3.5.2 Action Not Performed Due to Inadequate Procedure Adherence, was found to be a similar event. This event describes human performance errors related to briefings and procedure adherence that resulted in a TS violation.

G. COMPONENT FAILURE DATA

There was no component failure associated with this event.

This event has been reported to IRIS.